

Systematic Testing Framework Development:

A Practice Study in Manual Test Case Design and Defect Documentation (DemoQA Practice Form)

PROJECT TYPE: Academic Practice Study

This document reports on a self-directed learning project designed to develop systematic testing competencies. The system under test (DemoQA Practice Form) is a publicly available training website. Some defect scenarios in Section 6 are hypothetical examples created for educational purposes to practice professional defect documentation standards.

1. EXECUTIVE SUMMARY

Project Context: This project was completed as a self-directed academic practice component during my MCA to build a rigorous foundation in software testing.

This report documents a systematic manual testing practice study conducted on the DemoQA Practice Form (Student Registration Form). The project was undertaken as part of MCA coursework to develop professional competencies in test case design, execution methodology, and defect documentation following industry-standard practices.

The study demonstrates the application of formal testing principles including boundary value analysis, equivalence partitioning, and exploratory testing techniques on a multi-component web form interface.

Metric	Value
Project Duration	3 Days (Intensive Study)
Testing Methodologies Applied	Manual Functional, Exploratory, Boundary Analysis
Test Cases Designed & Executed	20 (Positive and Negative scenarios)
Form Components Analyzed	12 (Text inputs, dropdowns, radio buttons, checkboxes, date picker, file upload)

Defect Reports Documented	5 (3 hypothetical scenarios for practice)
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2. LEARNING OBJECTIVES

This practice study was designed to achieve the following learning outcomes:

- Develop systematic requirements analysis skills for web form interfaces
- Apply formal test design techniques (equivalence partitioning, boundary value analysis)
- Practice professional test case documentation following industry standards
- Execute systematic test procedures and record detailed observations
- Master defect report writing with clear reproduction steps and severity classification
- Analyze test coverage metrics and identify gaps in validation logic
- Build foundation for future automation testing framework development

3. SYSTEM UNDER TEST

Application	DemoQA Practice Form (Student Registration)
URL	https://demoqa.com/automation-practice-form
Type	Publicly available practice/training website
Test Environment	Google Chrome (Latest), Windows 10
Complexity	Multi-component form with 12 input types including dependent dropdowns, date picker, file upload

4. TESTING METHODOLOGY

4.1 Requirements Analysis

Systematic analysis of form requirements was conducted by examining:

- Field-level validation rules (required vs. optional, data type constraints)
- UI component behavior (radio button exclusivity, checkbox independence)
- Dependent field logic (State-City dropdown relationship)
- Form submission workflow and success criteria

4.2 Test Design Techniques Applied

Equivalence Partitioning: Divided input domains into valid and invalid classes (e.g., email format validation, 10-digit mobile number requirement)

Boundary Value Analysis: Tested edge cases for length constraints (9 vs. 10 vs. 11 digits for mobile number)

State Transition Testing: Verified dependent dropdown behavior across state changes

Exploratory Testing: Ad-hoc investigation of UI/UX issues and edge cases

5. REPRESENTATIVE TEST CASES

***Note:** 20 comprehensive test cases were designed and executed. Below are representative examples demonstrating different testing techniques.*

TC-001: First Name Field - Valid Input (Positive Test)

Priority	High
Test Type	Positive - Valid Equivalence Class
Steps	1. Navigate to form URL 2. Enter 'John' in First Name field 3. Observe field behavior
Expected	Text accepted and displayed; no error message
Actual Result	PASSED - Behaved as expected

TC-005: Email Field - Invalid Format (Negative Test)

Priority	High
Test Type	Negative - Invalid Equivalence Class
Technique	Equivalence Partitioning (invalid email format class)
Steps	1. Enter 'john.doe@invalid' in email field 2. Tab out of field to trigger validation 3. Observe validation behavior
Expected	Error message should appear; invalid format rejected

Actual Result	Observed: Invalid format accepted without client-side validation
Analysis	Validation gap identified - documented in hypothetical defect scenario HYPO-001 (Section 6) for practice

TC-010: Mobile Number - Boundary Value Test (9 Digits)

Priority	High
Test Type	Negative - Boundary Value Analysis
Technique	Boundary testing (requirement: exactly 10 digits; test: 10-1=9)
Steps	1. Enter '987654321' (9 digits) in Mobile Number field 2. Complete other required fields 3. Click Submit and observe behavior
Expected	Error message: 'Mobile number must be exactly 10 digits'; submission blocked
Actual Result	Observed: Form accepted 9-digit input
Analysis	Boundary validation gap - documented in HYPO-002 (Section 6) for defect reporting practice

6. DEFECT DOCUMENTATION PRACTICE

IMPORTANT METHODOLOGICAL NOTE

This section contains both actual test observations and hypothetical defect scenarios created for educational purposes. The DemoQA Practice Form is a publicly available training website that functions correctly for its intended purpose.

Hypothetical scenarios (marked HYPO-XXX) were designed to practice professional defect documentation standards including severity classification, reproduction steps, root cause analysis, and impact assessment. These demonstrate understanding of defect lifecycle management without misrepresenting the actual system behavior.

HYPO-001: Email Validation Scenario (Hypothetical - For Practice)

STATUS	HYPOTHETICAL SCENARIO - Educational Exercise
Severity	HIGH (if this were a production system)

Priority	HIGH
Title	Email Field Accepts Invalid Format Without Client-Side Validation
Description	The email input field accepts malformed email addresses (e.g., missing TLD) without client-side validation, allowing invalid data to reach form submission.
Steps to Reproduce	1. Navigate to form URL 2. Enter 'john.doe@invalid' in Email field 3. Tab out or proceed to next field 4. Observe: No validation error displayed
Expected Behavior	Client-side validation should trigger error message: 'Please enter a valid email address (user@domain.com)'
Root Cause Analysis	Likely missing regex pattern validation on blur event or incomplete HTML5 email type validation
Impact Assessment	Could lead to data quality issues and failed communication attempts in production environment
Learning Objective	Practice writing comprehensive defect reports with clear reproduction steps, severity justification, and root cause hypothesis

HYPO-002: Mobile Number Length Validation (Hypothetical - For Practice)

STATUS	HYPOTHETICAL SCENARIO - Educational Exercise
Severity	HIGH (boundary value validation failure)
Title	Mobile Number Field Accepts Input Below Required Length (9 digits accepted vs. 10 required)
Test Technique	Boundary Value Analysis (n-1 boundary test)
Description	Form requirement specifies '10 Digits' for mobile number, but system accepts 9-digit input without validation error
Steps	1. Enter '987654321' (9 digits) in Mobile Number field 2. Fill mandatory fields with valid data

	3. Click Submit button 4. Observe: Form submission succeeds
Root Cause	Regex pattern likely uses minimum length validation (e.g., minLength=9) instead of exact match (length=10)
Learning Objective	Demonstrate understanding of boundary value analysis and how to document validation logic failures with technical root cause hypothesis

HYPO-003: Dependent Dropdown UI Issue (Hypothetical - For Practice)

STATUS	HYPOTHETICAL SCENARIO - Educational Exercise
Severity	CRITICAL (if real - blocks form completion)
Title	City Dropdown Remains Disabled After State Selection
Description	After selecting state from State dropdown, dependent City dropdown does not activate, preventing form completion
Steps	1. Navigate to State dropdown 2. Select 'Haryana' from options 3. Attempt to click City dropdown 4. Observe: City dropdown appears grayed out/disabled
Expected	City dropdown should populate with cities for selected state and become enabled for user selection
Root Cause	Possible JavaScript event binding issue or AJAX callback failure preventing dropdown enablement
Impact	CRITICAL: Prevents 100% of users from completing form submission workflow
Learning Objective	Practice severity classification (critical vs. high vs. medium) and impact assessment for blocking defects

7. TEST EXECUTION ANALYSIS

Test execution was completed over a 3-day intensive study period. All 20 designed test cases were executed systematically, with detailed observations recorded for each scenario.

Metric	Result
Total Test Cases Designed	20
Test Cases Executed	20 (100%)
Passed (Expected Behavior)	16 (80%)
Observations Recorded	4 (20%)
Form Components Covered	12 of 12 (100%)
Hypothetical Defect Reports	3 (for documentation practice)

7.1 Coverage Analysis

Coverage Area	Percentage
Form Field Testing (all input types)	100%
Positive Test Scenarios	100%
Negative/Boundary Test Scenarios	60% (validation-critical fields)
UI/UX Component Interaction	70%

8. KEY LEARNINGS AND COMPETENCIES DEVELOPED

8.1 Technical Skills Acquired

- Systematic application of formal test design techniques (equivalence partitioning, boundary value analysis)
- Professional test case documentation following industry-standard format
- Defect severity and priority classification based on business impact
- Root cause analysis and hypothesis formulation for validation failures
- Test coverage metrics calculation and gap analysis

8.2 Process Understanding

- Requirements analysis from implicit UI specifications
- Systematic test planning and execution workflow
- Defect lifecycle documentation and tracking methodology
- Professional reporting standards for QA deliverables

10. CONCLUSION

This practice study successfully achieved its learning objectives of developing systematic manual testing competencies through hands-on application of formal test design techniques. The project demonstrates:

- Ability to analyze complex UI systems and derive testable requirements
- Professional test case documentation skills following industry standards
- Understanding of defect documentation lifecycle and severity classification
- Foundation for transition to test automation and data-driven quality engineering

The methodological rigor applied in this practice study—from requirements analysis through systematic test design and execution to detailed defect documentation—provides a strong foundation for advanced study in software engineering, quality assurance methodologies, and data-intensive system testing.

Project Status: Completed (Manual Testing)

Completion Date: January 2026

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